

Class 13: Transcriptomics and RNA-Seq Analysis

Your Name

Setup

```
library(DESeq2)
library(ggplot2)
library(dplyr)
library(AnnotationDbi)
library(org.Hs.eg.db)
```

Import Data

```
counts <- read.csv("airway_scaledcounts.csv", row.names=1)
metadata <- read.csv("airway_metadata.csv")
head(counts)
```

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	723	486	904	445	1170
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	523	616	371	582
ENSG000000000457	347	258	364	237	318
ENSG000000000460	96	81	73	66	118
ENSG000000000938	0	0	1	0	2
	SRR1039517	SRR1039520	SRR1039521		
ENSG000000000003	1097	806	604		
ENSG000000000005	0	0	0		
ENSG000000000419	781	417	509		
ENSG000000000457	447	330	324		

ENSG00000000460	94	102	74
ENSG00000000938	0	0	0

```
head(metadata)
```

	id	dex	celltype	geo_id
1	SRR1039508	control	N61311	GSM1275862
2	SRR1039509	treated	N61311	GSM1275863
3	SRR1039512	control	N052611	GSM1275866
4	SRR1039513	treated	N052611	GSM1275867
5	SRR1039516	control	N080611	GSM1275870
6	SRR1039517	treated	N080611	GSM1275871

Q1. How many genes are in this dataset?

```
nrow(counts)
```

```
[1] 38694
```

There are 38694 genes in this dataset.

Q2. How many 'control' cell lines do we have?

```
sum(metadata$dex == "control")
```

```
[1] 4
```

There are 4 control cell lines.

Toy Differential Gene Expression

```
control <- metadata[metadata[, "dex"]=="control",]  
control.counts <- counts[, control$id]  
control.mean <- rowSums(control.counts) / nrow(control)  
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          900.75          0.00          520.50          339.75          97.25  
ENSG000000000938  
          0.75
```

Q3. How would you make the above code more robust?

```
control <- metadata[metadata[, "dex"]=="control",]  
control.counts <- counts[, control$id]  
control.mean <- rowSums(control.counts) / nrow(control)  
head(control.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          900.75          0.00          520.50          339.75          97.25  
ENSG000000000938  
          0.75
```

Instead of hardcoding /4 we use `nrow(control)` to automatically count the number of control samples. This makes the code more robust if the number of samples changes.

Q4. Calculate the mean per gene for treated samples

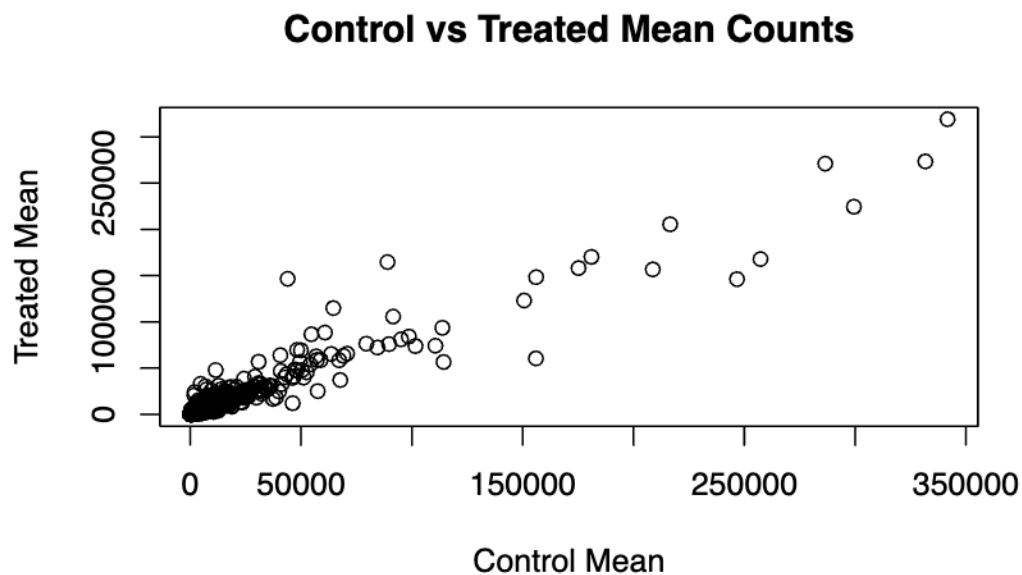
```
treated <- metadata[metadata[, "dex"]=="treated",]  
treated.counts <- counts[, treated$id]  
treated.mean <- rowSums(treated.counts) / nrow(treated)  
head(treated.mean)
```

```
ENSG000000000003 ENSG000000000005 ENSG000000000419 ENSG000000000457 ENSG000000000460  
          658.00          0.00          546.00          316.50          78.75  
ENSG000000000938  
          0.00
```

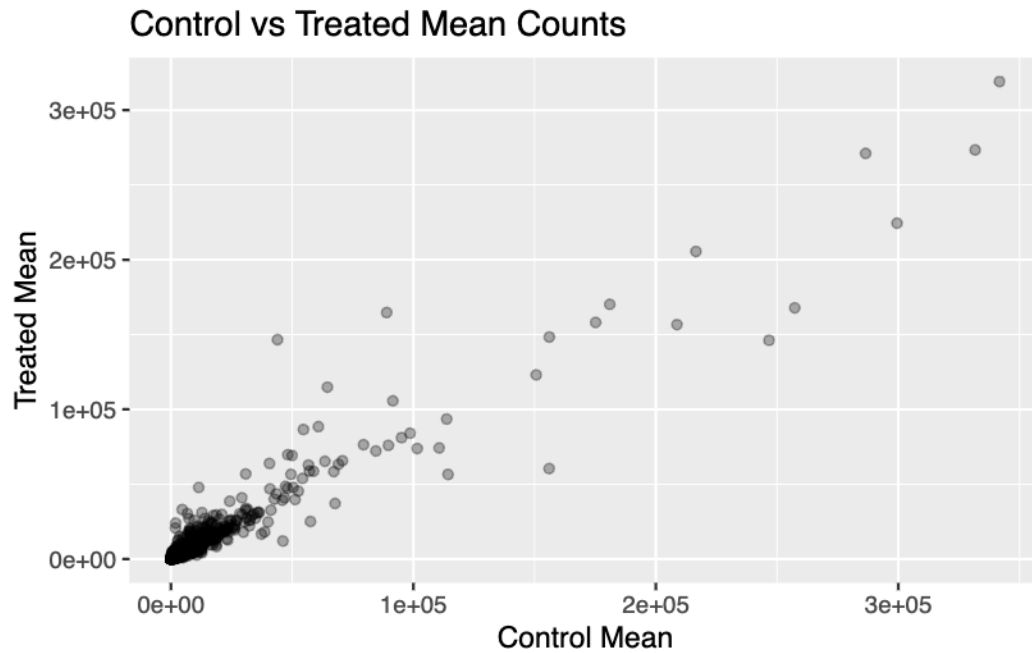
Q5. Scatter plot of control vs treated means

```
meancounts <- data.frame(control.mean, treated.mean)

plot(meancounts[,1], meancounts[,2],
     xlab="Control Mean",
     ylab="Treated Mean",
     main="Control vs Treated Mean Counts")
```



```
ggplot(meancounts, aes(x=control.mean, y=treated.mean)) +
  geom_point(alpha=0.3) +
  xlab("Control Mean") +
  ylab("Treated Mean") +
  ggtitle("Control vs Treated Mean Counts")
```

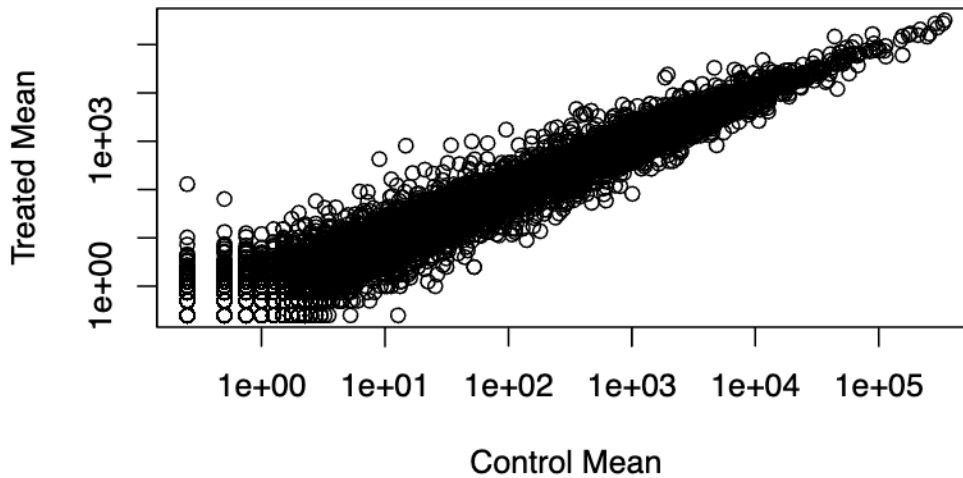


For Q5b I would use `geom_point()` for this scatter plot.

Q6. Plot both axes on a log scale

```
plot(meancounts[,1], meancounts[,2],  
     xlab="Control Mean",  
     ylab="Treated Mean",  
     log="xy",  
     main="Control vs Treated (log scale)")
```

Control vs Treated (log scale)



The argument `log="xy"` in the base R `plot()` function sets both axes to log scale. Using log scale reveals the spread of lowly expressed genes that were all clustered near the origin in the original plot.

Log2 Fold Change

```
meancounts$log2fc <- log2(meancounts[, "treated.mean"]/meancounts[, "control.mean"])
head(meancounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000005	0.00	0.00	NaN
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000938	0.75	0.00	-Inf

Q7. Purpose of `arr.ind` argument in `which()`

```
zero.vals <- which(meancounts[,1:2]==0, arr.ind=TRUE)
to.rm <- unique(zero.vals[,1])
mycounts <- meancounts[-to.rm,]
head(mycounts)
```

	control.mean	treated.mean	log2fc
ENSG000000000003	900.75	658.00	-0.45303916
ENSG000000000419	520.50	546.00	0.06900279
ENSG000000000457	339.75	316.50	-0.10226805
ENSG000000000460	97.25	78.75	-0.30441833
ENSG000000000971	5219.00	6687.50	0.35769358
ENSG00000001036	2327.00	1785.75	-0.38194109

The `arr.ind=TRUE` argument causes `which()` to return both the row AND column indices of TRUE values, rather than just a single element position number. This is important because we are checking a 2-column data frame for zeros and need to know which rows contain zeros. We take the first column (row indices) and call `unique()` to avoid trying to remove the same row twice in case a gene has zeros in both the control and treated columns.

Q8. How many up-regulated genes at $\log_2FC > 2$?

```
up.ind <- mycounts$log2fc > 2
sum(up.ind)
```

```
[1] 250
```

There are 250 up-regulated genes at a log2 fold change greater than 2.

Q9. How many down-regulated genes at $\log_2FC < -2$?

```
down.ind <- mycounts$log2fc < (-2)
sum(down.ind)
```

```
[1] 367
```

There are 367 down-regulated genes at a log2 fold change less than -2.

Q10. Do you trust these results?

No, these results should not be fully trusted. While we can see large fold changes, we have not performed any statistical testing to determine whether these differences are significant. We have no p-values or adjusted p-values, and we have not accounted for variability between replicates. A proper differential expression analysis using DESeq2 is needed.

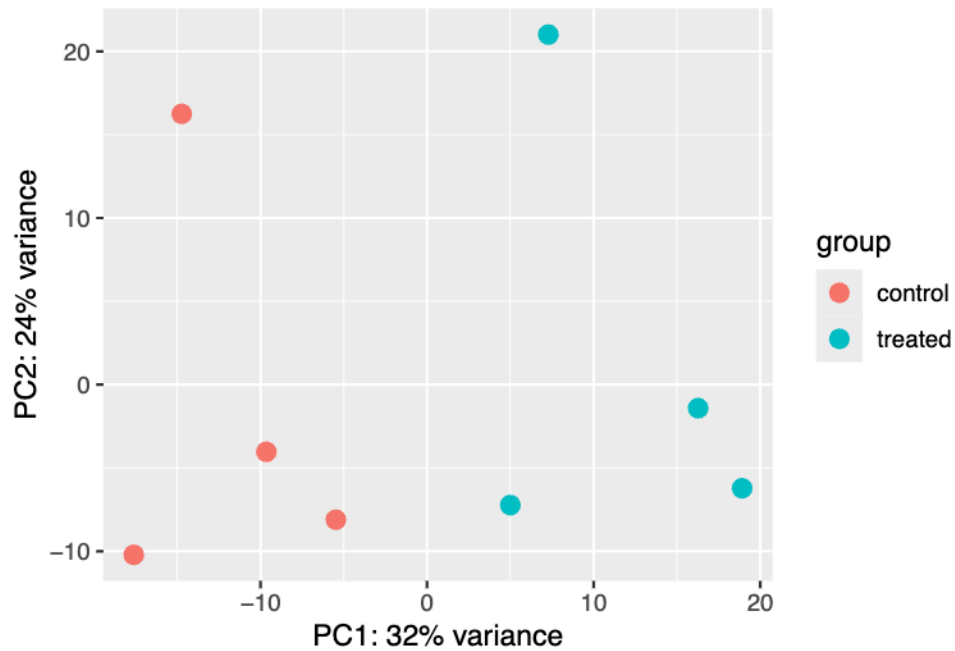
Setting Up DESeq2

```
dds <- DESeqDataSetFromMatrix(countData=counts,  
                              colData=metadata,  
                              design=~dex)  
dds
```

```
class: DESeqDataSet  
dim: 38694 8  
metadata(1): version  
assays(1): counts  
rownames(38694): ENSG000000000003 ENSG000000000005 ... ENSG00000283120  
               ENSG00000283123  
rowData names(0):  
colnames(8): SRR1039508 SRR1039509 ... SRR1039520 SRR1039521  
colData names(4): id dex celltype geo_id
```

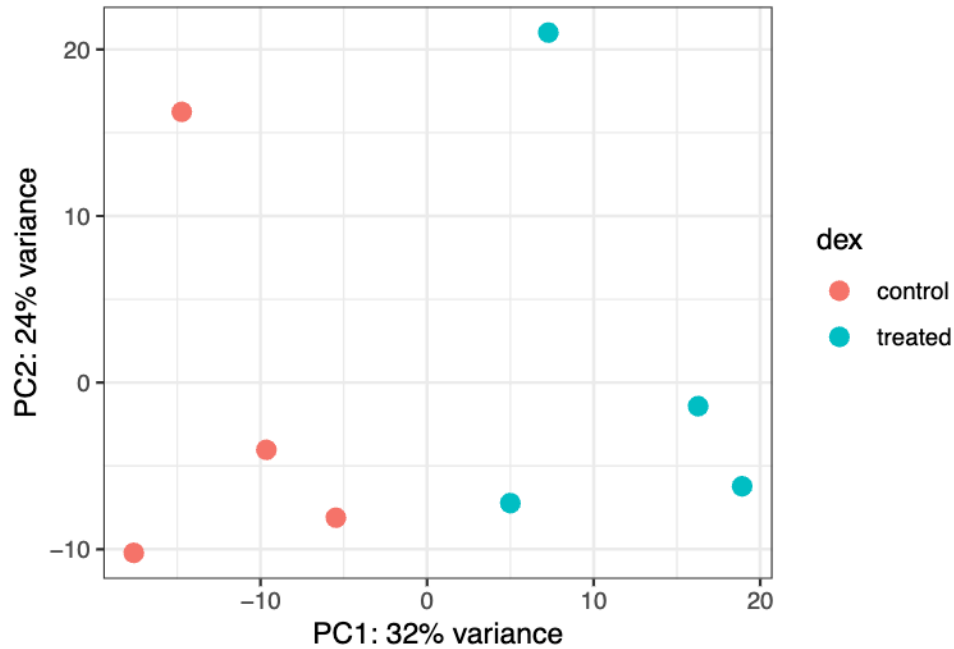
Principal Component Analysis (PCA)

```
vsd <- vst(dds, blind=FALSE)  
plotPCA(vsd, intgroup=c("dex"))
```



```
pcaData <- plotPCA(vsd, intgroup=c("dex"), returnData=TRUE)
percentVar <- round(100 * attr(pcaData, "percentVar"))

ggplot(pcaData) +
  aes(x=PC1, y=PC2, color=dex) +
  geom_point(size=3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  coord_fixed() +
  theme_bw()
```



DESeq Analysis

```
dds <- DESeq(dds)
res <- results(dds)
res
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 38694 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.1942	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.0000	NA	NA	NA	NA
ENSG000000000419	520.1342	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.6648	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.6826	-0.147143	0.256995	-0.572550	0.5669497
...	...	...	...	...	...
ENSG00000283115	0.000000	NA	NA	NA	NA
ENSG00000283116	0.000000	NA	NA	NA	NA
ENSG00000283119	0.000000	NA	NA	NA	NA
ENSG00000283120	0.974916	-0.66825	1.69441	-0.394385	0.693297

ENSG00000283123	0.000000	NA	NA	NA	NA
	padj				
	<numeric>				
ENSG00000000003	0.163017				
ENSG00000000005	NA				
ENSG000000000419	0.175937				
ENSG000000000457	0.961682				
ENSG000000000460	0.815805				
...	...				
ENSG00000283115	NA				
ENSG00000283116	NA				
ENSG00000283119	NA				
ENSG00000283120	NA				
ENSG00000283123	NA				

```
summary(res)
```

```
out of 25258 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 1564, 6.2%
LFC < 0 (down)    : 1188, 4.7%
outliers [1]      : 142, 0.56%
low counts [2]    : 9971, 39%
(mean count < 10)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

```
res05 <- results(dds, alpha=0.05)
summary(res05)
```

```
out of 25258 with nonzero total read count
adjusted p-value < 0.05
LFC > 0 (up)      : 1237, 4.9%
LFC < 0 (down)    : 933, 3.7%
outliers [1]      : 142, 0.56%
low counts [2]    : 9033, 36%
(mean count < 6)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Q11. Adding Annotation Data

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="SYMBOL",
                     multiVals="first")

res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")

res$uniprot <- mapIds(org.Hs.eg.db,
                      keys=row.names(res),
                      keytype="ENSEMBL",
                      column="UNIPROT",
                      multiVals="first")

res$genename <- mapIds(org.Hs.eg.db,
                       keys=row.names(res),
                       keytype="ENSEMBL",
                       column="GENENAME",
                       multiVals="first")

head(res)
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG000000000003	747.194195	-0.350703	0.168242	-2.084514	0.0371134
ENSG000000000005	0.000000	NA	NA	NA	NA
ENSG000000000419	520.134160	0.206107	0.101042	2.039828	0.0413675
ENSG000000000457	322.664844	0.024527	0.145134	0.168996	0.8658000
ENSG000000000460	87.682625	-0.147143	0.256995	-0.572550	0.5669497
ENSG000000000938	0.319167	-1.732289	3.493601	-0.495846	0.6200029
	padj	symbol	entrez	uniprot	
	<numeric>	<character>	<character>	<character>	

ENSG000000000003	0.163017	TSPAN6	7105	AOA087WYV6
ENSG000000000005	NA	TNMD	64102	Q9H2S6
ENSG000000000419	0.175937	DPM1	8813	H0Y368
ENSG000000000457	0.961682	SCYL3	57147	X6RHX1
ENSG000000000460	0.815805	FIRRM	55732	A6NFP1
ENSG000000000938	NA	FGR	2268	B7Z6W7

```

      genename
      <character>
ENSG000000000003      tetraspanin 6
ENSG000000000005      tenomodulin
ENSG000000000419 dolichyl-phosphate m..
ENSG000000000457 SCY1 like pseudokina..
ENSG000000000460 FIGNL1 interacting r..
ENSG000000000938 FGR proto-oncogene, ..

```

```
ord <- order(res$padj)
head(res[ord,])
```

log2 fold change (MLE): dex treated vs control

Wald test p-value: dex treated vs control

DataFrame with 6 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000152583	954.771	4.36836	0.2371306	18.4217	8.79214e-76
ENSG00000179094	743.253	2.86389	0.1755659	16.3123	8.06568e-60
ENSG00000116584	2277.913	-1.03470	0.0650826	-15.8983	6.51317e-57
ENSG00000189221	2383.754	3.34154	0.2124091	15.7316	9.17960e-56
ENSG00000120129	3440.704	2.96521	0.2036978	14.5569	5.27883e-48
ENSG00000148175	13493.920	1.42717	0.1003811	14.2175	7.13625e-46

	padj	symbol	entrez	uniprot
	<numeric>	<character>	<character>	<character>
ENSG00000152583	1.33157e-71	SPARCL1	8404	B4E2Z0
ENSG00000179094	6.10774e-56	PER1	5187	A2I2P6
ENSG00000116584	3.28806e-53	ARHGEF2	9181	AOA8Q3SIN5
ENSG00000189221	3.47563e-52	MAOA	4128	B4DF46
ENSG00000120129	1.59896e-44	DUSP1	1843	B4DRR4
ENSG00000148175	1.80131e-42	STOM	2040	F8VSL7

```

      genename
      <character>
ENSG00000152583      SPARC like 1
ENSG00000179094 period circadian reg..
ENSG00000116584 Rho/Rac guanine nucl..

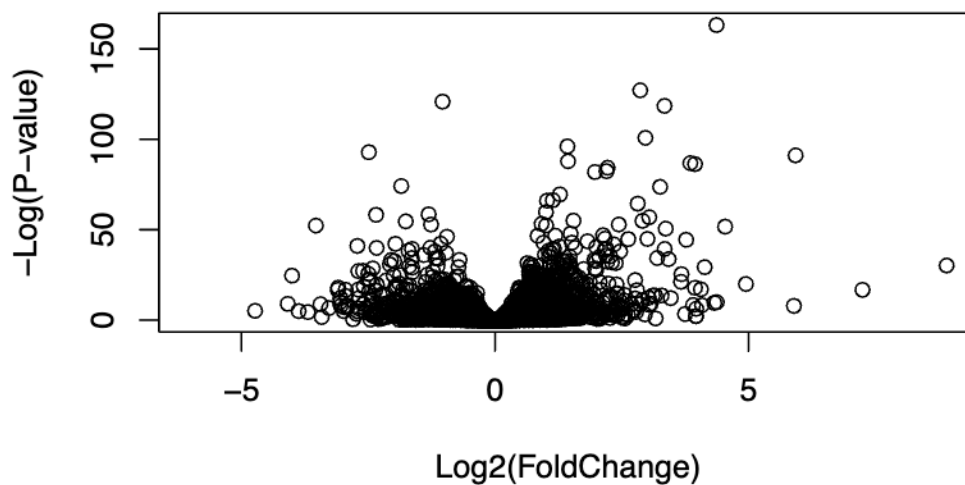
```

```
ENSG00000189221    monoamine oxidase A
ENSG00000120129 dual specificity pho..
ENSG00000148175      stomatin
```

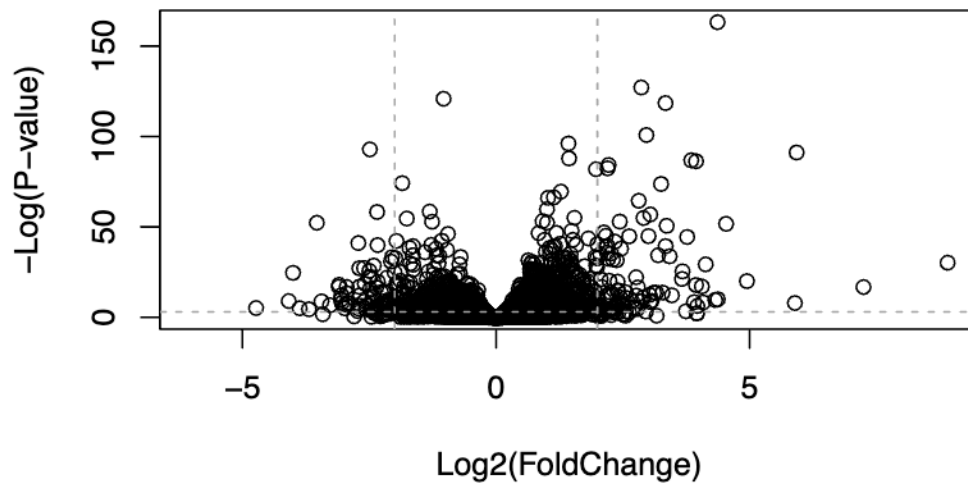
```
write.csv(res[ord,], "deseq_results.csv")
```

Volcano Plots

```
plot(res$log2FoldChange, -log(res$padj),
     xlab="Log2(FoldChange)",
     ylab="-Log(P-value)")
```

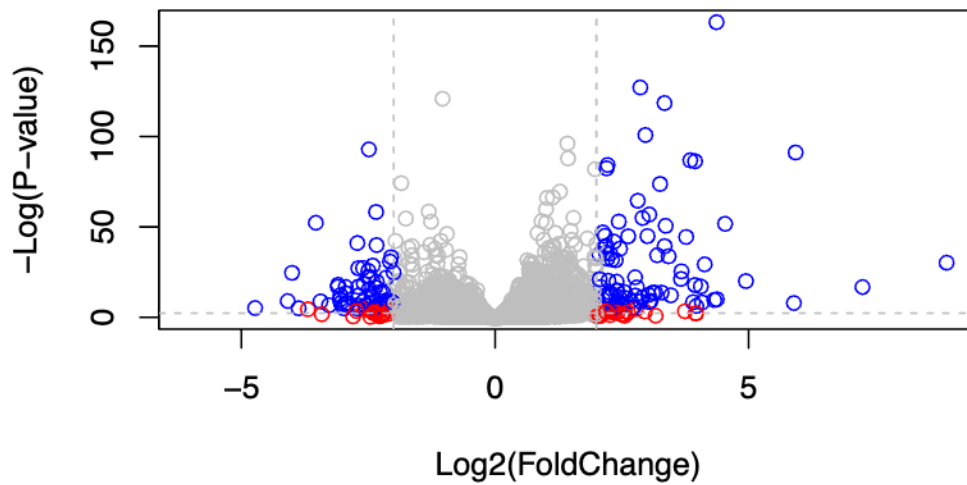


```
plot(res$log2FoldChange, -log(res$padj),
     ylab="-Log(P-value)", xlab="Log2(FoldChange)")
abline(v=c(-2,2), col="darkgray", lty=2)
abline(h=-log(0.05), col="darkgray", lty=2)
```



```
mycols <- rep("gray", nrow(res))
mycols[ abs(res$log2FoldChange) > 2 ] <- "red"
inds <- (res$padj < 0.01) & (abs(res$log2FoldChange) > 2)
mycols[ inds ] <- "blue"

plot(res$log2FoldChange, -log(res$padj),
      col=mycols, ylab="-Log(P-value)", xlab="Log2(FoldChange)")
abline(v=c(-2,2), col="gray", lty=2)
abline(h=-log(0.1), col="gray", lty=2)
```



Q12. Pathway Analysis - Top 2 Down-regulated Pathways

```
library(pathview)
library(gage)
library(gageData)

data(kegg.sets.hs)

foldchanges <- res$log2FoldChange
names(foldchanges) <- res$entrez

keggres <- gage(foldchanges, gsets=kegg.sets.hs)
head(keggres$less, 3)
```

		p.geomean	stat.mean	p.val
hsa05332	Graft-versus-host disease	0.0004250607	-3.473335	0.0004250607
hsa04940	Type I diabetes mellitus	0.0017820379	-3.002350	0.0017820379
hsa05310	Asthma	0.0020046180	-3.009045	0.0020046180

		q.val	set.size	exp1
hsa05332	Graft-versus-host disease	0.09053792	40	0.0004250607
hsa04940	Type I diabetes mellitus	0.14232788	42	0.0017820379

hsa05310 Asthma 0.14232788 29 0.0020046180

```
pathview(gene.data=foldchanges, pathway.id="hsa05332")
```

```
pathview(gene.data=foldchanges, pathway.id="hsa04940")
```

The top 2 down-regulated KEGG pathways are Graft-versus-host disease (hsa05332) and Type I diabetes mellitus (hsa04940). Both are immune-related pathways, consistent with the known anti-inflammatory effects of dexamethasone.

Session Information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Los_Angeles
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats4      stats      graphics  grDevices  utils      datasets  methods
[8] base
```

```
other attached packages:
```

```
[1] gageData_2.48.0      gage_2.60.0
[3] pathview_1.50.0      org.Hs.eg.db_3.22.0
[5] AnnotationDbi_1.72.0 dplyr_1.2.1
[7] ggplot2_4.0.2        DESeq2_1.50.2
[9] SummarizedExperiment_1.40.0 Biobase_2.70.0
[11] MatrixGenerics_1.22.0 matrixStats_1.5.0
```

[13]	GenomicRanges_1.62.1	Seqinfo_1.0.0
[15]	IRanges_2.44.0	S4Vectors_0.48.1
[17]	BiocGenerics_0.56.0	generics_0.1.4

loaded via a namespace (and not attached):

[1]	KEGGREST_1.50.0	gtable_0.3.6	xfun_0.57
[4]	lattice_0.22-9	bitops_1.0-9	vctrs_0.7.2
[7]	tools_4.5.3	parallel_4.5.3	tibble_3.3.1
[10]	RSQLite_3.52.0	blob_1.3.0	pkgconfig_2.0.3
[13]	Matrix_1.7-4	RColorBrewer_1.1-3	S7_0.2.1
[16]	graph_1.88.1	lifecycle_1.0.5	compiler_4.5.3
[19]	farver_2.1.2	Biostrings_2.78.0	codetools_0.2-20
[22]	htmltools_0.5.9	RCurl_1.98-1.18	yaml_2.3.12
[25]	GO.db_3.22.0	pillar_1.11.1	crayon_1.5.3
[28]	BiocParallel_1.44.0	DelayedArray_0.36.1	cachem_1.1.0
[31]	abind_1.4-8	tidyselect_1.2.1	locfit_1.5-9.12
[34]	digest_0.6.39	labeling_0.4.3	fastmap_1.2.0
[37]	grid_4.5.3	cli_3.6.5	SparseArray_1.10.10
[40]	magrittr_2.0.5	S4Arrays_1.10.1	XML_3.99-0.23
[43]	withr_3.0.2	scales_1.4.0	bit64_4.6.0-1
[46]	rmarkdown_2.31	XVector_0.50.0	httr_1.4.8
[49]	bit_4.6.0	otel_0.2.0	png_0.1-9
[52]	memoise_2.0.1	evaluate_1.0.5	knitr_1.51
[55]	rlang_1.1.7	Rcpp_1.1.1	glue_1.8.0
[58]	DBI_1.3.0	Rgraphviz_2.54.0	KEGGgraph_1.70.0
[61]	rstudioapi_0.18.0	jsonlite_2.0.0	R6_2.6.1